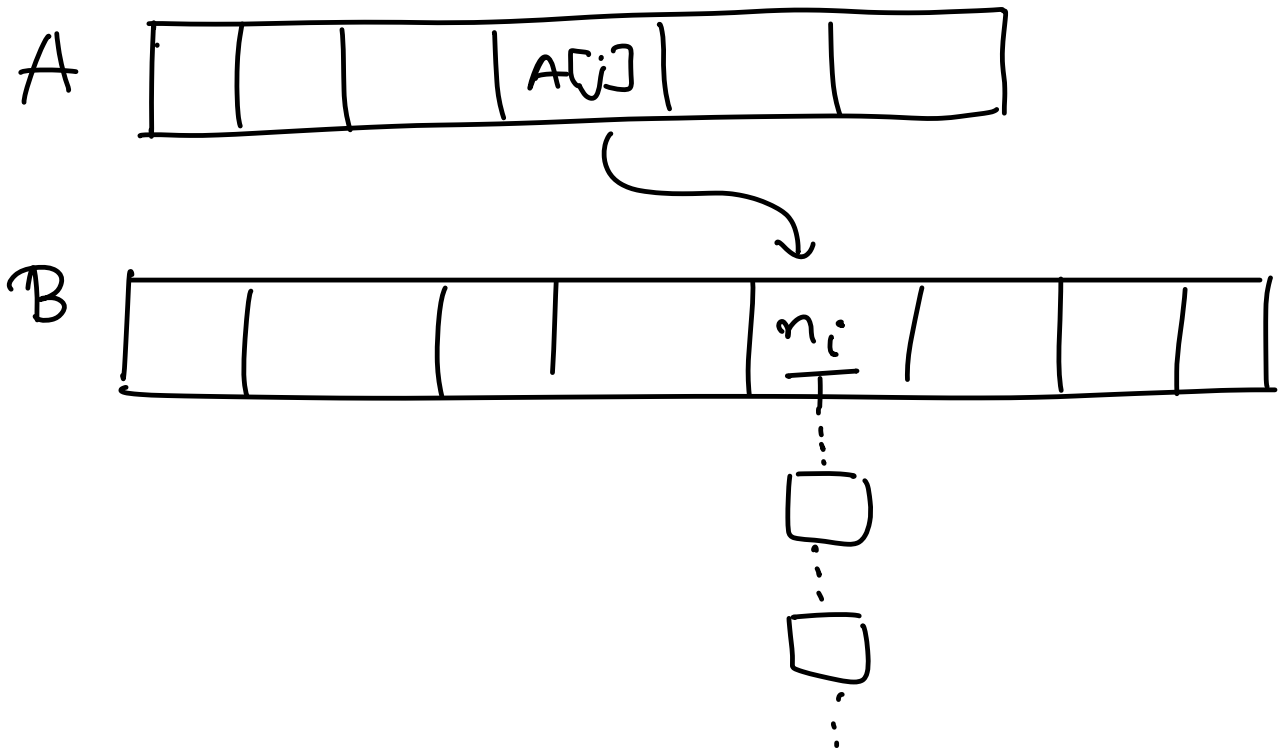
Bucket Sort:



$n_i = |B[i]|$

$$X_{ij} = \begin{cases} 1 & A[j] \text{ falls in } B[i]. \\ 0 \end{cases}$$

$$n_i = \sum_{j=1}^{n} X_{ij}$$

Time Complexity of Bucket Sort :

$$T(n) = \sum_{i=1}^{n} O(n_i^2) + \theta(n)$$

Average Case Analysis:

$$E(T(n)) = \sum_{i=1}^{n} O\left(E(n_i^2)\right) + \Theta(n)$$

$$= \sum_{i=1}^{n} O\left(E\left(\left(\sum x_{ij}\right)^2\right)\right) + \Theta(n) \quad \text{— (1)}$$

$$E(n_i^2) = E\left(\sum_{j=1}^{n} X_{ij}\right)^2$$

$$= E\left(\sum_{j=1}^{n} X_{ij}^2 + \sum_{\substack{j=1 \\ j\neq k}}^{n}\sum_{k=1}^{n} X_{ij}X_{ik}\right)$$

$$= \sum_{j=1}^{n} E(X_{ij}^2) + \sum_{j=1}^{n}\sum_{k=1}^{n} E(X_{ij})E(X_{ik})$$

$$= \sum_{j=1}^{n} \frac{1}{n} + \sum_{j=1}^{n} \sum_{k=1}^{n} \frac{1}{n} \cdot \frac{1}{n}$$

$$= \frac{1}{n} \sum_{j=1}^{n} 1 + \frac{1}{n^2} \sum_{\substack{j=1 \\ j \neq k}}^{n} \sum_{k=1}^{n} 1$$

$$= \frac{1}{n}(n) + \frac{1}{n^2}\left(n^2 - n\right)$$

$$= \frac{1}{n}(n) + \frac{1}{n^2}\left(n(n-1)\right)$$

$\sum_{j=1}^{n}\sum_{k=1}^{n} 1 = n^2$ ($j=k$ 'n' times)

$$= 1 + \frac{1}{n}(n-1)$$

$$= 1 + 1 - \frac{1}{n}$$

$$= 2 - \frac{1}{n}$$

$$\therefore$$

$$\boxed{E(n_i^2) = 2 - \frac{1}{n}}$$

Substituting in ①,

$$E(T(n)) = \sum_{i=1}^{n} O\left(2 - \frac{1}{n}\right) + \Theta(n)$$

$$= O\left(2 - \frac{1}{n}\right) \sum_{i=1}^{n} 1 + \Theta(n)$$

$$= n \cdot O\left(2 - \frac{1}{n}\right) + \Theta(n)$$

$$= \Theta(1) + \Theta(n)$$

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$= \Theta(n)$

$2 \to$ constant

$0 < \frac{1}{n} < 1$

So, $\Theta(n)$ dominates $n \cdot O\left(2+\frac{1}{n}\right)$ is small

$O(1) \leftrightarrow$ like cancels outs

Hence, Bucketsort runs in $\Theta(n)$ time on an average

Linear-Time Sorting Conditional Algorithms

A - input array.

Algorithms	Condition
Radix Sort (Counting Sort)	$k = \text{Range}[A] \in \Theta(n)$
Bucket Sort	A[i]'s are Uniformly distributed.

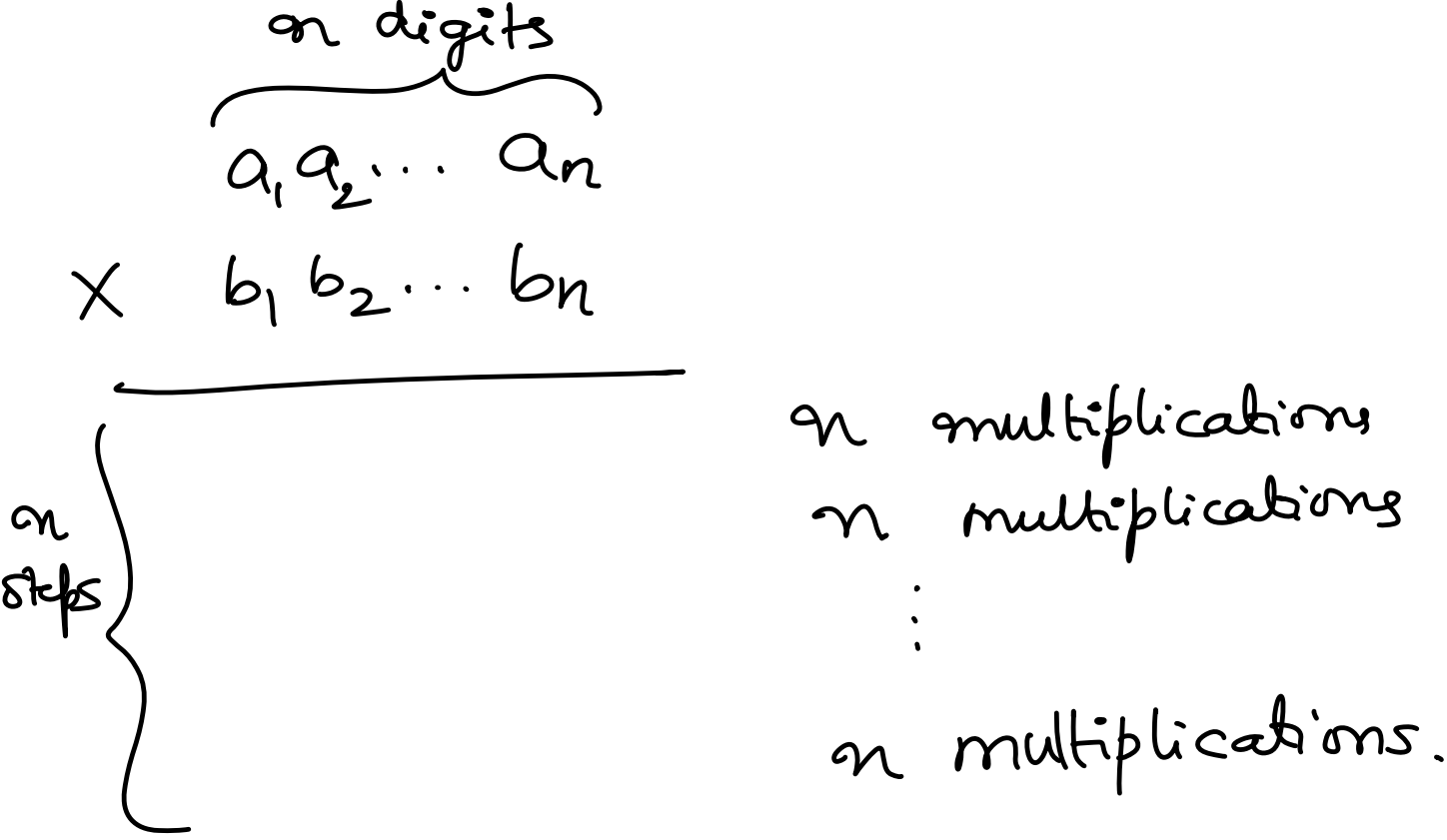
Integer Multiplication:

$$
\begin{array}{r}
34 \\
\times\ 21 \\
\hline
34 \\
68 \\
\hline
714
\end{array}
$$

→ 2 multiplications

→ 2 "

→ 1 Addition.



$\therefore$ Total multiplications $= n \cdot n = n^2$

Integer Multiplication using 3 multiplications: (of 2 digit numbers)

Write

$$34 = (3 \times 10^1) + (4 \times 10^0)$$

$$21 = (2 \times 10^1) + (1 \times 10^0)$$

$$(34) * (21) = (3 \times 10^1 + 4 \times 10^0) * (2 \times 10^1 + 1 \times 10^0)$$

$$= (3*2)\,10^2 + ((3*1) + (4*2))\,10^1 + (4*1)\,10^0$$

Idea:

Given $3*2$, $4*1$ write

$3*1+4*2$ using given multiplications and one extra multiplication.

$$(a+b)*(c+d) = a*(c+d) + b*(c+d)$$

$$= \underbrace{a*c}_{\text{done}} + (a*d + b*c) + \underbrace{b*d}_{\text{done}}$$

$$(a*d+b*c) = (a+b)*(c+d) - a*c - b*d$$

(left Distributivity & abelian of $+$ in $\mathbb{Z}$)

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Eg:

$$\underset{a\ d\ \ b\ c}{(3*1+4*2)} = (3+4)*(2+1) - 3*2 - 4*1$$

$$\| \qquad = 7*3 - 6 - 4$$

$$\| \qquad = 21 - 10$$

$$= 11$$

Karastuba Algorithm:

(Fast Multiplication of large Integers.)

Suppose we want to multiply a, b

$$a = a_1 a_0 = a_1 10^{n/2} + a_0$$

$$b = b_1 b_0 = b_1 10^{n/2} + b_0$$

$$c = a * b = \left(a_1 10^{n/2} + a_0\right) * \left(b_1 10^{n/2} + b_0\right)$$

$$= (a_1 * b_1) 10^{n} + \left(a_1 * b_0 + a_0 * b_1\right) 10^{n/2} + a_0 * b_0 .$$

$$c = c_2 10^{n} + c_1 10^{n/2} + c_0$$

where $c_2 = a_1 * b_1$

$c_0 = a_0 * b_0$

and $c_1 = (a_1 + a_0) * (b_1 + b_0) - c_2 - c_1 .$

eg: 345×217

a_1 a_0

0345

$\times$ 0217

b_1 b_0

(Make n even)

$n/2 = 2$

$c_2 = a_1 * b_1 = 6$

$c_0 = a_0 * b_0 = 45 * 17 = 765$

(Karatsuba algorithm iteratively)

$c_1 = (a_1 + a_0) * (b_1 + b_0) - c_2 - c_1$

$= (03+45) * (02+17) - 6 - 765$

$= (48 * 19) - 6 - 765$

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$= 912 - 6 - 765$

$= 141$

$C = c_2 10^4 + c_1 10^2 + c_0$

$= 6 \times 10^4 + (141) 10^2 + 1615$

$= 74865$

345×217

$= 74865$